Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Forward deployed engineer with 8+ years building multifamily PropTech at Entrata - the data warehouse, centralization layer, and operational backbone for affordable housing operators and resident workflows at 100K+ scale. I know the operations, leasing, compliance, and resident financial data your enterprise clients run on from the inside. At Entrata, I was the de facto embedded engineer for every affordable housing compliance deployment - scoping requirements with customer stakeholders in ambiguous environments, building solutions end-to-end, integrating across dependent systems, and owning production outcomes. Now building production agentic AI: LeaseI is an 8-agent LangGraph pipeline for leasing compliance investigation (93.6% precision); AgentProbe ships a production eval harness validating it across 1,240+ test cases. I scope fast, build production-grade Python and JavaScript, and ship 0-to-1 systems without needing a roadmap handed to me. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Multifamily PropTech Domain Expertise | Customer-Embedded AI Deployment | LLM-Powered Resident Workflows | Agentic AI & LangGraph | Enterprise Data Centralization | 0-to-1 Product Ownership | HITL Design & AI Guardrails | Production Python & FastAPI

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML models and agentic AI pipelines; integrating data systems with back-end APIs; iterating on production behavior and team feedback in a fast-moving startup environment.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **Scoped customer needs in an ambiguous environment, designed agentic solution, built and shipped end-to-end:** translated fuzzy marketing and real estate survey requests into a LangGraph pipeline (parse → plan → retrieve → validate → conditional routing), delivering structured, compliant outputs automatically - primary technical contact across all stakeholders throughout delivery.
- Integrated RAG (FAA Part 107 corpus) for grounded compliance decisions; engineered guardrails blocking non-compliant outputs before reaching customers; brought learnings back to rapidly improve the pipeline post-launch.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated data integrity checks; designed Tableau/Quarto dashboards for faculty leadership (+40% comprehension).
- Fine-tuned a custom classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; presented evaluation results to faculty stakeholders.

Entrata (PropTech / Multifamily Real Estate SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Acted as primary embedded technical contact for multifamily affordable housing operators** - scoped their compliance, operations, and resident financial requirements, built solutions end-to-end, integrated across 5+ dependent systems, and owned production outcomes for 100K+ monthly active users across 7 years. Zero compliance violations, zero client-impacting incidents traced to my modules.
- **Owned the centralized data pipeline for operations, leasing eligibility, and resident financial workflows:** annual income-limit imports (federal rows → 180,000 domain records), rent and HAP calculations, eligibility determinations - every decision auditable and compliant with HUD/LIHTC regulations. Built and maintained the RabbitMQ-based async processing system for gross-rent updates with idempotency guarantees.
- Designed and integrated RESTful APIs and microservices across internal platforms; led AWS S3 migration with security controls (Libsodium, 100% PHI compliance, zero incidents); eliminated 80% of manual operational workload via Python/Bash automation pipelines across 120+ Agile sprints.
- Optimized 200+ SQL queries (30% performance gain) on marketing, finance, and operations data; mentored 5+ engineers; drove technical standards via 300+ code reviews; built internal debugging tools cutting team debugging time by 40%.

PROJECTS

Leasel: Agentic Leasing & Compliance Investigation Platform

LangGraph 8-agent -- HITL -- FastAPI -- RAG -- Multimodal

Built a production agentic AI platform for the exact investigation workflow multifamily operators face: a resident's eligibility is disputed, evidence is scattered across lease documents, system settings, regulatory rules, and incident history - and manual investigation takes a full day. Three surfaces: **Investigation** (8-agent LangGraph pipeline producing ranked, evidence-backed root-cause hypotheses with HITL escalation when confidence too low); **Document Intelligence** (multimodal OCR + TF-IDF RAG, citation grounding on lease PDFs, human-review gate on regulated decisions); **Resident Self-Service** (AI answers policy questions; eligibility decisions always route to staff). Results: 93.6% precision / 87.9% F1 on 1,240 lease PDFs; 100% citation correctness on Phase 1 eval set.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: AI Agent Eval Harness & Orchestration Toolkit

LangGraph -- LangChain -- FastAPI -- ChromaDB -- MCP

Production eval harness for agentic AI systems - 15-endpoint FastAPI backend, 5-metric LLM-as-judge engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI/CD regression gate blocking failing runs automatically, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), generative AI prompt engineering pipeline, observability dashboard. Used to validate LeaseI across 1,240+ compliance test cases. Ships production-grade Python; open-source on GitHub.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP-Driven Clinical Records Query System

Flask -- LLMs -- BERT -- React -- Docker -- FHIR

Scoped with clinical staff, designed, and deployed an NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC; reduced manual retrieval time ~60%; drove staff adoption across clinical workflows as primary technical contact through go-live.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, Generative AI, LLMs, Generative Models, Multi-Agent Orchestration, Intent Classification, Tool Integration, Multi-Step Reasoning, HITL, AI Guardrails, Safety Controls, Prompt Engineering, RAG Pipelines, Semantic Search, Embeddings, LLM Routing, OpenAI API, Anthropic API, FastAPI, Observability, Hallucination Detection, MCP, 0-to-1 Product Ownership

Languages & Frameworks: Python, JavaScript, TypeScript, Node.js, Java, SQL, React.js, Django, Spring Boot, Flask, Streamlit, PHP

Cloud & Infra: AWS (S3, EC2, Lambda, Glue, Bedrock), Docker, Kubernetes, GitHub Actions CI/CD, PostgreSQL, RabbitMQ, Redis, Microservices, Distributed Systems, Data Warehouse Design, Semantic Layer

ML & Data: PyTorch, HuggingFace Transformers, scikit-learn, XGBoost, Isolation Forest, Pandas, NumPy, Tableau, Operations Data, Finance Data, Resident Workflow Automation